

# Gengcong Yang

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## EDUCATION

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| <b>University of Texas at Austin (UT Austin)</b><br><i>Master in Computer Science</i>                                              | 08/2023 – 12/2024<br>Austin, United States |
| <b>Tsinghua University (THU)</b><br><i>Master in Control Engineering, Department of Automation   Supervisor: <u>Yujiu Yang</u></i> | 08/2019 – 06/2022<br>Beijing, China        |
| <b>National Chiao Tung University (NCTU)</b><br><i>Exchange Student in Computer Science</i>                                        | 09/2017 – 01/2018<br>Hsinchu, Taiwan       |
| <b>Sun Yat-sen University (SYSU)</b><br><i>Bachelor in Software Engineering</i>                                                    | 08/2015 – 06/2019<br>Guangzhou, China      |

## PUBLICATIONS

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| <b>Diamond: Harnessing GPU Resources for Scientific Deep Learning</b>                                                                                                                                                                                                                                                                                                        | 2025 |
| <ul style="list-style-type: none"><li>Authors: Haotian Xie, Rohan Marwaha, Minu Mathew, Song Bian, <b>Gengcong Yang</b>, et al.</li><li>Contributed to the design of a distributed job-submission pipeline enabling scalable, multi-node GPU training on a platform for the scientific machine learning lifecycle; funded by <b>NSF</b> and published in eScience.</li></ul> |      |
| <b>Probabilistic Modeling of Semantic Ambiguity for Scene Graph Generation</b>                                                                                                                                                                                                                                                                                               | 2021 |
| <ul style="list-style-type: none"><li>Authors: <b>Gengcong Yang</b>, Jingyi Zhang, Yong Zhang, Baoyuan Wu, Yujiu Yang</li><li>Proposed a probabilistic framework and designed a plug-and-play uncertainty modeling module to capture semantic ambiguity and multiple plausible visual relationships in scene graphs.</li><li>Published in <b>CVPR 2021</b>.</li></ul>        |      |

## RESEARCH EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                   |                             |
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| <b>Texas Advanced Computing Center, Graduate Research Assistant</b>                                                                                                                                                                                                                                                                                                                                                               | Austin, 01/2024 – 12/2024   |
| <ul style="list-style-type: none"><li>Supervisor: <u>Yinzhi Wang</u></li><li>Contributed to the early-stage development of <u>Diamond</u>, an NSF-funded, scalable platform supporting the full machine learning research lifecycle on HPC resources.</li><li>Designed and implemented a distributed job-submission pipeline enabling users to containerize environments and train ML models across multiple GPU nodes.</li></ul> |                             |
| <b>KAUST, Research Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                     | Remote, 02/2021 – 07/2021   |
| <ul style="list-style-type: none"><li>Supervisor: <u>Bernard Ghanem</u></li><li>Implemented a differentiable RoI Pooling method with CUDA and C++ for the temporal action detection task on videos, achieving an experimental improvement of 6%.</li></ul>                                                                                                                                                                        |                             |
| <b>Tencent AI Lab, Computer Vision Researcher Intern</b>                                                                                                                                                                                                                                                                                                                                                                          | Shenzhen, 07/2019 – 07/2020 |
| <ul style="list-style-type: none"><li>Supervisor: <u>Baoyuan Wu</u></li><li>Developed a novel modeling method for polysemous visual-relationship representation and applied for a technology patent.</li><li>Contributed to the design and experimentation of the probabilistic uncertainty modeling framework later published at CVPR 2021.</li></ul>                                                                            |                             |

## INDUSTRY EXPERIENCE

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**TikTok USDS**, Emerging Product, *Machine Learning Engineer* San Jose, 07/2026 – Present

- Led safety guardrails to detect and block unsafe prompts and generated videos for Seedance, a video generation model.
- Built and scaled GPU model-serving infrastructure for U.S. deployment.
- Developed LLM-based agents that automate the team's operational workflows, reducing manual effort.

**Uber**, Driver Offer Pricing Team, *Machine Learning Engineer* Sunnyvale, 01/2025 – 07/2026

- Developed driver fare prediction models used as core pricing signals within Uber's marketplace pricing system.
- Improved observability and reliability infrastructure for pricing prediction pipelines running at scale.

**ByteDance**, E-commerce, *Machine Learning Engineer* Shanghai, 06/2022 – 06/2023

- Developed image search models and a retrieval-ranking framework for the TikTok Shop similar-product search project.
- Constructed large-scale datasets for post-pretraining and quantized object detection models to improve serving efficiency.

**Tencent**, *Machine Learning Engineering Intern* Beijing, 06/2021 – 08/2021

- Focused on the retrieval stage of the recommendation system for an early-stage short-video platform.
- Built an offline evaluation pipeline for retrieval models and experimented with pairwise training for a two-tower model.

## PATENTS

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### Visual Relationship Recognition Method Based on Artificial Intelligence

- Inventors: **Gengcong Yang**, Yong Zhang, Baoyuan Wu, Yanbo Fan, Zhifeng Li, Yujiu Yang, Wei Liu
- Patent Number: CN202011108484
- Granted Date: 03/05/2024

## SELECTED AWARDS

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|---------------------------------------------------------|------------|
| Tsinghua SIGS Professional Practice Scholarship         | 2022       |
| Tsinghua Excellent Scholarship First Prize              | 2021       |
| SYSU Outstanding Undergraduate Thesis Award             | 2019       |
| SYSU Excellent Scholarship Second Prize                 | 2018       |
| National Encouragement Scholarship                      | 2017, 2018 |
| SYSU Xian Weijian Overseas Study Funding                | 2017       |
| Interdisciplinary Contest In Modeling Honorable Mention | 2017       |
| SYSU Excellent Scholarship Third Prize                  | 2016, 2017 |